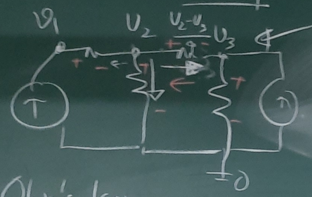
$\frac{v_2-v_1}{1}$

Recap



Nodal analysis ← KCL / variables = pot of nodes

Curr. source

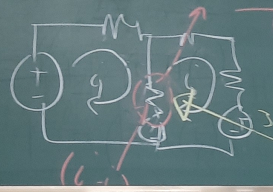
Mesh " → KVL / var. mesh current

Volt. source

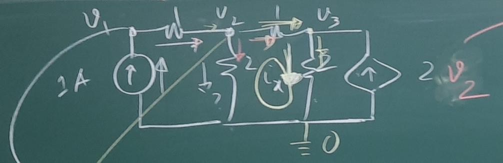
Ohm's law

Comm

↳ eff curr.



$i_2 - i_1$



VCCS

CCCS

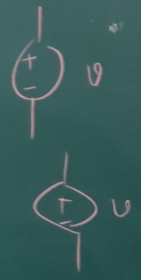
$$\frac{v_1-v_2}{1} = 1$$

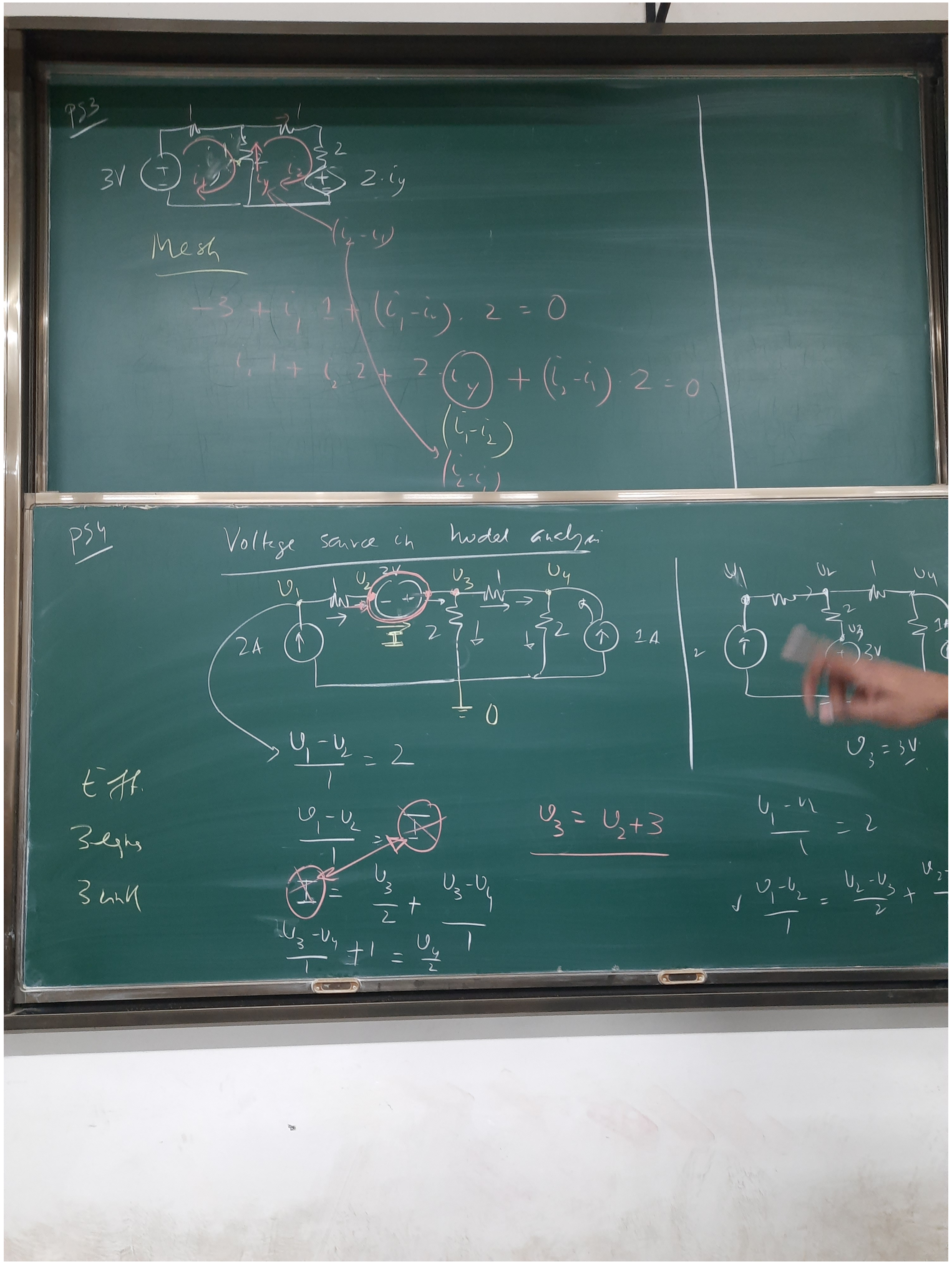
$$\frac{v_1-v_2}{1} = \frac{v_2-v_3}{1} + \frac{v_2-0}{2}$$

$$\frac{v_3}{2} = \frac{v_2-v_3}{1} + 2\,\underbrace{i_x}_{}\quad i_x = \frac{v_3}{2}$$

$v_1 \quad v_2 \quad v_3$

PS2





PS3

Mesh

$$-3 + i_1 \cdot 1 + (i_1 - i_2)\cdot 2 = 0$$

$$i_1 \cdot 1 + i_2 \cdot 2 + 2 \cdot \boxed{i_y} + (i_2 - i_1)\cdot 2 = 0$$

$(i_1 - i_2)$

$(i_2 - i_1)$

PS4

Voltage source in nodal analysis

$$\frac{v_1 - v_2}{1} = 2$$

t̄ ff.

3 eqns

3 unk

$$\frac{v_1 - v_2}{1} = \boxed{\cancel{I}}$$

$$\boxed{\cancel{I}} = \frac{v_3}{2} + \frac{v_3 - v_4}{1}$$

$$\frac{v_3 - v_4}{1} + 1 = \frac{v_4}{2}$$

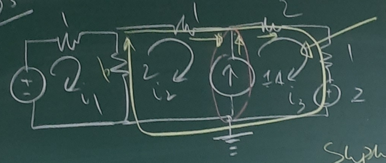
$$v_3 = v_2 + 3$$

$$v_3 = 3V$$

$$\frac{v_1 - v_2}{1} = 2$$

✓ $\frac{v_1 - v_2}{1} = \frac{v_2 - v_3}{2} + v_2 -$

ps5



$i_3 = i_2 + CS$

Eff

2 eq

2 unk

Supr-mesh

$$i_2 \cdot 1 + \underline{\overset{(i_2+cs)}{i_3} \cdot 2} + i_3 \cdot 1 + 2 + 2 \cdot (i_2 - i_1) = 0$$

$$-3 + i_1 \cdot 1 + \underbrace{(i_1 - i_2) \cdot 2}_{} = 0$$

Approach

Count nodes / mesh

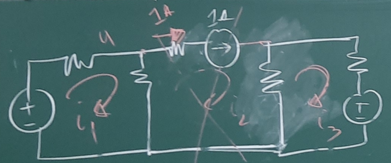
Count CS / VS

Nodal → eff eq = #nodes − #VS

Mesh anc

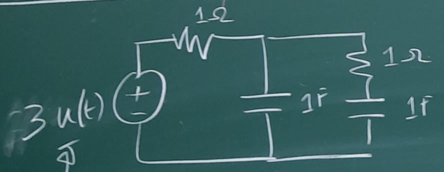
#eqn = #mesh − #CS

ps6



$$-\frac{3}{s} + i_1(s) + (i_1(s) - i_2(s)) \cdot \frac{1}{s} + \frac{V(0^+)}{s} = 0$$

$$+ \left((i_2(s) - i_1(s))\right)\frac{1}{s}$$



unit step